Applied ML

26-02-26

☆ Bag of words approach

text → tokenization → stop words removal → lemmatization or stemming → document frequency table

food is good	+ve
drinks are really great	+ve
desserts are very very good	+ve
service is very bad	-ve

↓

tokenization → stop words removal → lemmatization

vocab = {food, good, drink, real, great, dessert, very, serve, bad}

Document Term Frequency Table

doc	food	good	drink	real	great	dessert	very	serve	bad
1	1	1	0	0	0	0	0	0	0
2	0	0	1	1	1	0	0	0	0
3	0	1	0	0	0	1	1	0	0
4	0	0	0	0	0	0	1	1	1

food is good = [1 1 0 0 0 0 0 0 0]

text ⟶ vector (numerical representation of text)

text → vector → $\boxed{\text{ML model}}$ → prediction target

Issues with BOW approach :

1) Frequency of words in the text is not taken into account.

2) Exclusive words are not given any special treatment.

eg.

desserts are very very good

↓

[dessert, very, good] BOW

the token 'very' is present twice, but its not taken into account here.

solution for this is creating Term Frequency for every word.

	frequency of word within given document	Term Frequency
dessert	1	1/4
very	2	2/4
good	1	1/4
total	4	

for term / token = t and document = d

$$TF(t, d) = \frac{\text{frequency of token} = t \text{ in document} = d}{\text{no. of total tokens in document} = d}$$

[dessert, very, good] = $\left[\frac{1}{4} \quad \frac{2}{4} \quad \frac{1}{4}\right]$

vector

eg.

1) Battery is great. Screen is really great

↓ text preprocessing

	TF
battery	1/5
great	2/5
screen	1/5
real	1/5

2) phone is great.

↓ text preprocessing

	TF
phone	1/2
great	1/2

vocab = {battery, great, screen, real, phone}

doc	battery	great	screen	real	phone
1	1/5	2/5	1/5	1/5	0
2	0	1/2	0	0	1/2

this captures the length of text as well as the frequency of words.

this is better than simple BOW approach

eg.

food is good	+ve
service was nice	+ve
dessert is good	+ve
parking was bad	-ve

good - 2 docs - very common in +ve reviews
nice - 1 doc - very exclusive in +ve reviews

compared to 'good' 'nice' is more exclusive

- we need to capture the exclusiveness of word 'nice'

solution for this is Inverse Document Frequency

$$IDF(t) = \log\left(\frac{\text{no. of total documents in corpus}}{\text{no. of documents in which term=t is present}}\right)$$

$$idf(good) = \log\left(\frac{4}{2}\right) = \log(2)$$

$$idf(nice) = \log\left(\frac{4}{1}\right) = \log(4)$$

$$idf(nice) > idf(good)$$

this is how more importance is given to more exclusive term

food is good	+ve
service was nice	+ve
dessert is good	+ve
parking was bad	-ve

vocab = { food, good, serve, nice, dessert, park, bad }

Term Frequencies

doc	food	good	serve	nice	dessert	park	bad
1	$\frac{1}{2}$	$1/2$	0	0	0	0	0
2	0	0	$1/2$	$1/2$	0	0	0
3	0	$1/2$	0	0	$1/2$	0	0
4	0	0	0	0	0	$1/2$	$1/2$

IDF

doc	food	good	serve	nice	dessert	park	bad
1	$\log(\frac{4}{1})$	$\log(\frac{4}{2})$	0	0	0	0	0
2	0	0	$\log(\frac{4}{1})$	$\log(\frac{4}{1})$	0	0	0
3	0	$\log(\frac{4}{1})$	0	0	$\log(\frac{4}{1})$	0	0
4	0	0	0	0	0	$\log(\frac{4}{1})$	$\log(\frac{4}{1})$

final representation of token = TF × IDF

food is good → [food, good]

$$\begin{array}{c} \left[\ \frac{1}{2}\times\log\left(\frac{4}{1}\right) \quad \frac{1}{2}\times\log\left(\frac{4}{2}\right) \quad 0 \quad 0 \quad 0 \quad 0 \quad 0\ \right] \\ \text{food} \quad \text{good} \quad \text{serve} \quad \text{nice} \quad \text{dessert} \quad \text{park} \quad \text{bad} \end{array}$$

← final vector representation

tf-idf = tf × idf

tf → captured the frequency within the document

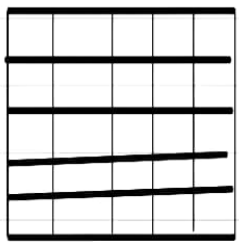
idf → exclusiveness within the corpus

Image Data

images are made up of pixels

↓ 50 pixels

50 × 100 → 100 pixels



5

5

every pixel has 3 LED bulbs

↓

$\underbrace{\text{Red, Green, Blue}}_{\text{base colors}}$

Red Green Blue LEDs have intensity betn 0-255

by increasing or decreasing the intensities we can generate any visible color.

Every image stores pixel intensities for these 3 colors

every colorful image has 3 channels → red, green, blue

every black-white image has single channel → gray scale

If the size of image is 50×100 then it stores pixel intensity values of R, G, B for all 50×100 pixels.

total values stored in this image: $\underset{\substack{\downarrow \\ \text{3 channels RGB}}}{3} \times 50 \times 100$

shape of every colorful image = $3 \times h \times w$

which makes image a 3 dimensional array

R	G	B
0	0	0

↓ pure black

⟷

R	G	B
255	255	255

↓ pure white